FIRST SEMESTER M.Sc. DEGREE EXAMINATION, NOVEMBER 2018

(CUCSS—PG)

Mathematics

MT 1C 02—LINEAR ALGEBRA

(2016 Syllabus Year)

Time : Three Hours Maximum : 36 Weightage

Part A

Answer **all** *the questions.*
Each question carries weightage 1.

1. Let v be a vector space over a field F. Show that if c is a scalar and o is the zero-vector then $c \cdot 0 = 0$.

2. Let V be the set of all pairs (x, y) of real numbers, and let F be the field of real numbers. Define :

$$(x, y) + (x_1, y_1) = (x + x_1, 0)$$
$$c(x, y) = (cx, 0)$$

Is V with these operation a vector space ?

3. Let V be the vector space of all $n \times n$ matrices over the field real numbers. Is that the set of all non-inveritable matrices A form a subspace of V ?

4. Verify that the Transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ is defined by $T(x, y)\ ((x - 1) + (y - 1), 2x + 3y)$ is linear transformation or not.

5. Let $\mathcal{B}$ $\{(1, 0, -1), (1, 1, 1), (2, 2, 0)\}$ be a basis for C^3. Find the dual basis of $\mathcal{B}$.

6. Find the linear transformation on $\mathbb{R}^3$ which transform the vector (1, 5, 9) to (– 5, 1, – 9).

7. Show that similar matrices have the same characteristic polynomial.

8. Prove that every matrix A such that $A^2 = A$ is similar to diagonal matrix.

9. True *or* False : If a diagonalizable matrics has its only characteristic values zero and one, then it is a projection. Justify.

10. Let T be the linear operator on $\mathbb{R}^2$, the matrix of which in the standard ordered basis is $\begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$.

Let W_1 be the subspace of $\mathbb{R}^2$ spanned by (1, 0). Show that W_1 is invariant under T.

11. If N is a nilpotent operator on α an n-dimensional vector space V, then characteristic polynomial for N is x^n.

12. Let $\alpha = (x_1, x_2)$ and $\beta = (y_1, y_2)$ in $\mathbb{R}^2$. Check whether $\mathbb{R}^2$ with the product $(\alpha \mid \beta) = x_1y_1 - x_2y_1 - x_1y_2 + 4x_2y_2$ is an inner product space or not.

13. Show that if S is any subset of a vector space V, then its orthogonal complement $S^\perp$ is a subspace of V.

14. If V (F) be a vector space of polynomials int t with the product is defined by $\langle p(t), q(t) \rangle = \int_0^1 p(t)\, q(t)\, dt$. If $p(t) = t + 2$ then find $\|p\|^2$.

(14 × 1 = 14 weightage)

Part B

Answer any seven questions.
Each question carries weightage 2.

15. Find the co-ordinates of the vector (2, 1, – 6) of $\mathbb{R}^3$ relative to the basis $\alpha_1 = (1,1, 2)$, $\alpha_2 = (3, -1, 0)$, $\alpha_3 = (2, 0, -1)$.

16. Let V be the space of 2 × 2 matrices over F. Find a basis $\{A_1, A_2, A_3, A_4\}$ for V such that $A_j^2 = A_j$ for each j.

17. Let W_1 and W_2 be the subspace of a finite dimensional vector space V. Prove that $(W_1 \cap W_2)^\circ = W_1^\circ + W_2^\circ$.

18. If S is any subset of a finite dimensional vector space V, then prove that $(s^0)^0$ is the subspace spanned by S.

19. Let T be the linear operator on R^4 which is represented in the standard order matrix $\begin{pmatrix} 0 & 0 & 0 & 0 \\ a & 0 & 0 & 0 \\ 0 & b & 0 & 0 \\ 0 & 0 & c & 0 \end{pmatrix}$.

Under what condition on a, b and c is T diagonalizable ?

20. Let T be a diagonalizable linear operator on finite dimensional vector space V, and let W be a subspace which is invariant under T. Prove that the restriction operator T_W is diagonalizable.

21. If $W_1 = \{(x, 0, z) : x, z \in R\}$, $W_2 = \{(0, y, z) : y, z \in R\}$ be two subspaces of R^3, then prove that $R^3 = W_1 + W_2$ but not $R^3 = W_1 \oplus W_2$.

22. Let E be the projection and T be a linear operator on a vector space V. Show that the range and null space of E are invariant under T if and only if ET = TE.

23. Let S be a orthogonal set of non-zero vectors in an inner product space V. Show that S is a linearly independent set.

24. State and prove Bessel's inequality.

(7 × 2 = 14 weightage)

Part C

Answer any two questions.
Each question carries 4 weightage.

25. Let V be the vector space of all 2 × 2 matrices over the field. Let W_1 be the set of matrices of the form $\begin{pmatrix} x & -x \\ y & z \end{pmatrix}$ and let W_2 be that set of matrices of the form $\begin{pmatrix} a & b \\ -a & c \end{pmatrix}$:

(a) Prove that W_1 and W_2 are subspace of V.

(b) Find the dimensions of W_1, W_2, $W_1 + W_2$ and $W_1 \cap W_2$.

26. Let W be the subspace of R^5 spanned by the vectors $\alpha_1 = (2, -2, 3, 4 - 1)$, $\alpha_2 = (-1, 1, 2, 5, 2)$, $\alpha_3 = (0, 0, -1, -1, 3)$, $\alpha = (1, -1, 2, 3, 0)$. Describe W^0, the annihilator of W.

27. Let V be a fine-dimensional vector space over the field F and let T be a linear operator P on V. Then T is diagonalizable if and only if the minimal polynomial for T has the form $p(x) = (x - c_1)(x - c_2) \ldots (x - c_k)$, where $c_2, c_2 \ldots c_k$ are distinct elements of F.

28. Apply the Gram-Schmidt process to the vector process to the vector $\beta_1 = (3, 0, 4)$, $\beta_2 = (-1, 0, 7)$, ($\beta_3 = (2, 9, 11)$, to obtain an orthonormal basis for $\mathbb{R}^3$ with the standard inner product.

(2 × 4 = 8 weightage)